

TUTORIAL – 4: Functional & Non-Functional Testing

Aim

To apply **Functional Testing, Non-Functional Testing, Positive Testing, and Negative Testing** on the **Login, Registration, and Payment modules** of a sample application and verify whether the software behaves as expected under different testing conditions.

Description

Software Testing is performed to verify that the software works correctly and satisfies user requirements.

This practical covers four important testing techniques:

1. Functional Testing

Functional Testing verifies whether every feature of the software works according to the Software Requirement Specification (SRS).

Examples:

- Login with valid credentials.
- Register a new user.
- Complete payment successfully.

2. Non-Functional Testing

Non-Functional Testing checks system quality attributes instead of functionality.

It includes:

- Performance Testing
- Load Testing
- Stress Testing
- Security Testing
- Usability Testing
- Compatibility Testing

Example:

Check whether the Login page opens within 3 seconds.

3. Positive Testing

Positive Testing verifies the application using **valid inputs**.

Example:

Username : admin

Password : admin123

Expected Result:

Login Successful

4. Negative Testing

Negative Testing verifies the application using **invalid or unexpected inputs**.

Example:

Username : admin
Password : 123
Expected Result:
Invalid Username or Password

Difference

Functional Testing	Non-Functional Testing
Checks what the system does	Checks how the system performs
Based on requirements	Based on quality attributes
Verifies features	Verifies speed, security, usability
Example: Login	Example: Response Time

Problem Statement

Perform Functional, Non-Functional, Positive, and Negative Testing on the **Login Module** of the Student Management System. Design appropriate test cases, execute them, record observations, and verify whether the software satisfies the functional and quality requirements.

How to Perform this Practical (Student Procedure)

Step 1: Study the Requirements

Read the Software Requirement Specification (SRS) and identify the functionality of the Login module.

Requirement:

- User should log in using a valid username and password.
- Invalid credentials should display an error message.
- Login page should load within 3 seconds.

Step 2: Prepare Test Cases

Create different test cases for:

- Functional Testing
- Non-Functional Testing
- Positive Testing
- Negative Testing

Step 3: Execute the Test Cases

Run the application and perform testing using the following inputs.

Test	Input
Positive Test	admin / admin123
Negative Test	admin / 12345
Blank Test	Empty Username & Password
Performance Test	Measure Login Response Time

Step 4: Compare Results

Compare:

Expected Result

vs

Actual Result

If both are same

Status = PASS

Otherwise

Status = FAIL

Step 5: Record Observation

Write observations after execution.

Example:

- ✓ Login successful using valid credentials.
- ✓ Error message displayed for invalid credentials.
- ✓ Login page loaded in less than 2 seconds.

Implementation Details

The Student Management System provides secure login for registered users.

Testing performed:

- Functional Testing
- Non-Functional Testing
- Positive Testing
- Negative Testing

Parameter Details

Parameter	Details
Project Name	Student Management System
Application/Software	Student Management System
Module Name	Login Module
SDLC Model	Waterfall Model
Programming Language	Python
IDE/Tool Used	PyCharm / VS Code
Input/Test Data	Username: admin Password: admin123

Source Code

```
username = input("Enter Username : ")
password = input("Enter Password : ")

if username == "admin" and password == "admin123":
    print("Login Successful")
else:
    print("Invalid Username or Password")
```

Test Case Detailed

Field	Details
Test Case ID	TC001
Requirement ID	R001
Module Name	Login Module
Feature Under Test	Login Authentication
Test Scenario	Login with valid username and password

Test Objective	Verify successful login
Test Type (Unit/Integration/System/UAT)	System Testing
Test Technique (Black Box/White Box)	Black Box Testing
Priority	High
Severity	Critical
Preconditions	Application should be running
Postconditions	User enters Home Page
Test Environment	Windows 10
Browser/OS (if applicable)	Google Chrome / Windows 10
Test Data	admin / admin123
Expected Result	Login Successful
Actual Result	Login Successful
Status (Pass/Fail)	Pass
Executed By	Student Name
Execution Date	DD/MM/YYYY

Additional Test Cases

Test Case ID	Testing Type	Test Data	Expected Result	Status
TC001	Positive Testing	admin/admin123	Login Successful	Pass
TC002	Negative Testing	admin/12345	Invalid Password Message	Pass

TC003	Negative Testing	Blank Username	Username Required	Pass
TC004	Negative Testing	Blank Password	Password Required	Pass
TC005	Functional Testing	Valid Login	Home Page Opens	Pass
TC006	Non-Functional Testing	Response Time	Login Page opens within 3 sec	Pass

Test Case Execution

Step No.	Test Steps	Input	Expected Result	Actual Result	Status
1	Open Application	-	Login page displayed	Login page displayed	Pass
2	Enter Username	admin	Username accepted	Username accepted	Pass
3	Enter Password	admin123	Password accepted	Password accepted	Pass
4	Click Login	Click	Login Successful	Login Successful	Pass
5	Enter Wrong Password	12345	Error Message	Error Message Displayed	Pass

Testing Activities Performed

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1	Requirement Analysis	Requirement Analysis	Study SRS	Requirement Document
2	Design	Test Planning	Prepare Test Plan	Test Plan
3	Development	Test Case Design	Create Test Cases	Test Case Document

4	Testing	Test Execution	Execute Positive & Negative Tests	Test Execution Report
5	Testing	Test Closure	Prepare Final Test Report	Test Summary Report

Observation

Sr. No.	Observation	Remarks
1	Login successful with valid credentials	Pass
2	Invalid credentials rejected	Pass
3	Blank fields display validation messages	Pass
4	Login page response time is within 3 seconds	Pass
5	All functional requirements satisfied	Pass

Output

Enter Username : admin

Enter Password : admin123

Login Successful

Positive Testing : PASS

Negative Testing : PASS

Functional Testing : PASS

Non-Functional Testing : PASS

Solve the Questionnaire

1. Which testing verifies software functionality against requirements?

- A) Non-Functional Testing
- B) Functional Testing**
- C) Security Testing
- D) Performance Testing

Answer: B) Functional Testing

2. Positive Testing is performed using:

- A) Invalid Inputs
- B) Incorrect Data
- C) Valid Inputs**
- D) Random Inputs

Answer: C) Valid Inputs

3. Which testing verifies system behavior with invalid inputs?

- A) Positive Testing
- B) Regression Testing
- C) Integration Testing
- D) Negative Testing**

Answer: D) Negative Testing

4. Performance Testing is a type of:

- A) Functional Testing
- B) Non-Functional Testing**
- C) Unit Testing
- D) Acceptance Testing

Answer: B) Non-Functional Testing

5. Which module is commonly tested for valid username and password verification?

- A) Payment Module
- B) Registration Module
- C) Login Module**
- D) Billing Module

Answer: C) Login Module

Conclusion / Outcome

- Understood the concepts of **Functional Testing**, **Non-Functional Testing**, **Positive Testing**, and **Negative Testing**.

- Designed and executed test cases for the Login module using valid and invalid inputs.
- Verified that the application behaves correctly for both expected and unexpected user inputs.
- Evaluated the application's performance by checking the login response time.
- Learned how different testing techniques help ensure software quality, reliability, and user satisfaction.